# Prediction of Properties of Additively Manufactured Parts For Haynes 282

Undergraduate Project Report

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# Abstract

Laser Powder Bed Fusion (LPBF) is one of the most advanced additive manufacturing techniques for fabricating high-performance metallic components with complex geometries. However, achieving defect-free parts with optimal mechanical properties requires precise control of process parameters such as laser power, scan speed, hatch spacing, and layer thickness.

This project presents a machine learning-based framework for predicting the properties of LPBF-fabricated Haynes 282 alloy components using experimentally collected and literature-derived datasets. A dataset consisting of approximately 500 samples was developed to establish relationships between process parameters and output properties including relative density, melt pool dimensions, hardness, porosity, and defect classification.

Multiple ensemble learning algorithms, including CatBoost Regressor, XGBoost Regressor, and LightGBM Regressor, were implemented and evaluated. Among the tested models, CatBoost demonstrated the best predictive performance with high accuracy and low error metrics across multiple properties.

Additionally, a process parameter optimization interface was developed to recommend suitable LPBF parameters for achieving desired target properties. The proposed framework provides a computationally efficient alternative to expensive experimental trials and numerical simulations, thereby contributing toward intelligent and data-driven additive manufacturing.

# Chapter 1

# Introduction

## 1.1 Background

Additive Manufacturing (AM), commonly referred to as 3D printing, has emerged as a transformative manufacturing technology capable of fabricating complex geometries directly from digital models. Unlike conventional subtractive manufacturing processes, additive manufacturing builds components layer-by-layer, thereby reducing material wastage and enabling unprecedented design flexibility.

Among the various additive manufacturing techniques, Laser Powder Bed Fusion (LPBF) has gained significant attention for the fabrication of high-performance metallic components used in aerospace, automotive, biomedical, and energy industries. In LPBF, a high-energy laser selectively melts metallic powder particles layer-by-layer according to the geometry defined by a CAD model.

The quality and performance of LPBF-manufactured components are highly dependent on process parameters such as laser power, scan speed, hatch spacing, and layer thickness. Improper combinations of these parameters can lead to defects such as porosity, lack of fusion, keyholing, and poor surface finish.

## 1.2 Haynes 282 Alloy

Haynes 282 is a nickel-based superalloy widely used in high-temperature applications due to its excellent creep resistance, oxidation resistance, thermal stability, and weldability. The alloy is particularly suitable for aerospace turbine components and high-temperature structural applications.

The alloy primarily consists of nickel, chromium, cobalt, molybdenum, titanium, and aluminum. The presence of titanium and aluminum promotes gamma-prime precipitation strengthening, while chromium enhances oxidation resistance.

Despite its excellent mechanical properties, processing Haynes 282 through LPBF

presents several challenges due to the complex thermal behavior involved during rapid melting and solidification.

## 1.3 Need for Process Optimization

The LPBF process involves a large parameter space, making experimental optimization both time-consuming and expensive. Traditional trial-and-error methods require extensive experimentation to determine suitable processing conditions for achieving high relative density and desirable mechanical properties.

Machine learning techniques offer an efficient alternative by learning relationships between input process parameters and output material properties from experimental and literature-derived datasets. Predictive models can significantly reduce experimental efforts and accelerate process optimization.

## 1.4 Problem Statement

In LPBF manufacturing, achieving consistent part quality remains a major challenge due to the nonlinear interaction between process parameters and material behavior. Small variations in laser power, scan speed, hatch spacing, or layer thickness can drastically influence melt pool characteristics, porosity formation, and mechanical properties.

Therefore, there is a need to develop a robust predictive framework capable of estimating material properties and recommending optimized process parameters for defect-free manufacturing.

## 1.5 Objectives

The primary objectives of this project are as follows:

1. To develop a dataset combining experimental and literature-derived LPBF process data for Haynes 282 alloy.

2. To predict properties such as relative density, melt pool width, melt pool depth, hardness, and defect class using machine learning techniques.

3. To compare the performance of different ensemble learning models including CatBoost, XGBoost, and LightGBM.

4. To generate printability maps for identifying stable and defective process regions.

5. To develop an inverse optimization framework capable of recommending suitable LPBF process parameters for desired output properties.

## 1.6 Scope of the Work

This work focuses on LPBF fabrication of Haynes 282 alloy using experimentally collected and literature-derived datasets. The study primarily investigates the influence of four process parameters:

- Laser Power
- Scan Speed
- Hatch Spacing
- Layer Thickness

The developed machine learning framework aims to provide accurate prediction and optimization capabilities while minimizing the need for extensive physical experimentation.

# Chapter 2

# Literature Review

## 2.1 Introduction

Laser Powder Bed Fusion (LPBF) has emerged as one of the most widely adopted metal additive manufacturing techniques due to its capability to fabricate highly complex geometries with excellent dimensional accuracy and mechanical performance. The process involves selective melting of metallic powder using a high-energy laser beam in a layer-by-layer manner.

In recent years, significant research efforts have focused on understanding the influence of LPBF process parameters on melt pool dynamics, porosity formation, microstructure evolution, and final mechanical properties. Machine learning and data-driven modeling techniques are also increasingly being explored for process optimization and property prediction.

This chapter presents a review of existing literature related to LPBF processing of Haynes 282 alloy, process parameter optimization, melt pool behavior, defect formation, and machine learning applications in additive manufacturing.

## 2.2 Laser Powder Bed Fusion Process

LPBF is a powder-bed additive manufacturing process in which thin layers of metallic powder are selectively melted using a focused laser beam. The process repeats layer-by-layer until the complete three-dimensional geometry is fabricated.

The major process parameters influencing the LPBF process include:

- Laser Power
- Scan Speed
- Hatch Spacing

- Layer Thickness

The interaction between these parameters governs the thermal history, melt pool geometry, cooling rate, and solidification behavior of the material.

The volumetric energy density is commonly used to characterize the combined effect of LPBF process parameters:

$$E = \frac{P}{vht} \tag{2.1}$$

where:

- $E$ = Volumetric Energy Density
- $P$ = Laser Power
- $v$ = Scan Speed
- $h$ = Hatch Spacing
- $t$ = Layer Thickness

A suitable energy density range is critical for achieving dense and defect-free components.

## 2.3 Haynes 282 Alloy in Additive Manufacturing

Haynes 282 is a gamma-prime strengthened nickel-based superalloy developed for high-temperature structural applications, particularly in aerospace and gas turbine industries. The alloy exhibits excellent creep resistance, oxidation resistance, thermal stability, and weldability.

According to Haynes International, Haynes 282 possesses superior fabricability compared to conventional nickel-based superalloys while maintaining excellent high-temperature strength characteristics [1].

Shaikh et al. investigated the feasibility of manufacturing Haynes 282 using LPBF and demonstrated the successful fabrication of fully dense and crack-free components [2]. Their study reported that LPBF-fabricated Haynes 282 exhibited strong anisotropic grain structures due to directional solidification during layer-wise manufacturing.

The study further highlighted that the mechanical properties of LPBF-manufactured Haynes 282 at room temperature exceeded those of conventionally wrought material in certain conditions. However, anisotropy in tensile behavior and reduced ductility at elevated temperatures were observed.

Boswell et al. studied the influence of energy density and heat treatment on LPBF-manufactured Haynes 282 [3]. Their work demonstrated that heat treatment significantly reduced anisotropy by promoting recrystallization and equiaxed grain formation.

## 2.4 Defect Formation in LPBF

One of the major challenges in LPBF is the formation of defects caused by improper process parameter combinations. Common defects include:

- Lack of Fusion
- Keyholing
- Gas Porosity
- Balling
- Surface Irregularities

Lack of fusion defects occur when insufficient energy input prevents complete melting between adjacent tracks or layers. On the other hand, excessive energy input can lead to keyhole porosity caused by vaporization-induced cavity formation.

Khairallah et al. investigated pore formation mechanisms in LPBF and demonstrated the influence of recoil pressure, Marangoni convection, and melt pool instability on defect generation [4].

The study explained that stable melt pool formation is essential for minimizing porosity and ensuring uniform material consolidation.

## 2.5 Melt Pool Behavior

The melt pool is one of the most critical aspects of the LPBF process since it directly governs solidification behavior, microstructure evolution, and defect formation.

Melt pool dimensions such as width and depth are strongly influenced by laser power and scan speed. Higher laser power generally increases melt pool dimensions, whereas higher scan speed reduces interaction time and decreases melt pool size.

Several studies have shown that melt pool morphology can be used as an indicator of process stability and printability. Stable melt pools produce dense parts with reduced porosity, while unstable melt pools lead to irregular solidification and defect formation.

Recent advancements in in-situ monitoring and thermal imaging have enabled real-time observation of melt pool evolution during LPBF processing.

## 2.6 Machine Learning in Additive Manufacturing

Machine learning techniques are increasingly being adopted in additive manufacturing for process modeling, defect prediction, parameter optimization, and quality control.

Traditional finite element and physics-based simulations of LPBF are computationally expensive and time-consuming due to the complex thermal-fluid interactions involved during melting and solidification. Machine learning models provide a computationally efficient alternative by learning patterns directly from experimental and simulated datasets.

Various machine learning algorithms including Random Forest, Support Vector Machines, Artificial Neural Networks, Gradient Boosting, XGBoost, and CatBoost have been successfully applied for predicting material properties and process outcomes.

Liu et al. proposed a physics-informed machine learning framework for porosity prediction in LPBF [5]. Their study incorporated physical parameters such as energy density and recoil pressure into machine learning models to improve generalization capability.

Recent studies have also explored digital twin frameworks combining physics-based simulations and machine learning for real-time process optimization and defect monitoring.

## 2.7 Research Gap

Although significant research has been conducted on LPBF processing of nickel-based superalloys, several limitations still exist.

Most studies focus either on experimental characterization or isolated parameter optimization. Limited work has been reported on the development of unified datasets combining literature-derived and experimentally collected LPBF data for Haynes 282 alloy.

Furthermore, many studies primarily investigate individual output properties such as density or hardness, while comprehensive multi-property prediction frameworks remain limited.

There is also a need for inverse optimization systems capable of recommending process parameters for desired target properties. Such systems can significantly reduce experimental cost and accelerate industrial adoption of LPBF.

## 2.8 Summary

The literature review indicates that LPBF processing of Haynes 282 is a promising area of research due to the alloy's excellent high-temperature properties and printability characteristics. Previous studies have demonstrated the feasibility of producing dense and crack-free Haynes 282 components through careful process optimization.

Machine learning techniques have shown significant potential for process prediction and optimization in additive manufacturing. However, the development of generalized datasets, multi-property prediction frameworks, and inverse parameter recommendation systems remains an active research area.